

EZEKWE JOHNBOSCO

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Senior Software Engineer and Technical Lead with 10+ years of experience building national-scale health and fintech systems across Africa. Experienced in leading engineering teams, architecting distributed platforms, and delivering secure, high-volume transaction systems. Proven track record across government health deployments, HIV surveillance infrastructure, and financial platforms.

Key Competencies:

System Architecture • Distributed Systems • Health Informatics • Fintech Platforms • Cloud Infrastructure • API Design • API Development • Data Systems • Engineering Leadership

Career Highlights:

- Senior Software Engineer with 10+ years experience building scalable fintech and health information systems.
- Designed and deployed distributed data systems supporting national healthcare programs and large epidemiological studies.
- Led engineering teams delivering high-availability platforms used by governments and public health organizations.
- Built secure microservices and API-driven systems processing large volumes of health and financial data.
- Delivered systems operating across multiple African countries and national health infrastructures.

CORE TECHNOLOGIES

Languages: C#, Python, Ruby, TypeScript, JavaScript, VB.NET

Frameworks: ASP.NET Core, Django, Ruby on Rails, React, React Native

Databases: MSSQL, PostgreSQL, MySQL, Redis, Cosmos DB, MongoDB

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Nginx

Architecture: Microservices, Distributed Systems, REST APIs, Event-Driven Systems

Security: OAuth2, JWT, RBAC

Testing: Unit Testing, Integration Testing

PROFESSIONAL EXPERIENCE

Senior Software Engineer

EasyXchange LTD | Gaborone, Botswana

2024 – PRESENT (20 hours a week)

- Architected and deployed a Python/Django URL shortening platform handling high-volume redirects with Redis caching, achieving 99.9% uptime.
- Designed and built a multi-tenant fintech platform enabling suppliers to access early invoice payments. Implemented authentication, automated invoice workflows, and financial reporting.
- Leading development of a microservices e-wallet platform supporting secure transactions, fraud monitoring, and scalable payment processing.

Technologies: Django, Rails, PostgreSQL, Redis, Docker, AWS, Pytest, Rubocop.

Health Informatics Lead

BUMMHI | Gaborone, Botswana

2023 – PRESENT

- Leading development of the national Patient Information Management System (PIMS) deployed across 500+ healthcare facilities for the Botswana Ministry of Health. Architected modules for patient records, reporting, SMS reminders, and facility-level health data management.
- Designed StudySpace, a real-time monitoring platform used in national HIV PHIA surveys to coordinate 300+ field teams and track survey progress.
- Architected a nationwide laboratory proficiency testing platform used by public and private laboratories across Botswana.

Technologies: C# ASP.NET, VB.NET, MSSQL, MySQL, AWS.

Regional Software Development Manager

University of Maryland | Baltimore, USA

2021 – 2023

- Led engineering for a COVID-19 cohort monitoring system supporting 25,000+ participants and 14+ follow-ups per participant, integrated with WHO's VigiFlow vaccine safety platform.
- Designed national seroprevalence study systems covering 3,400+ households, enabling real-time data validation and reporting.
- Architected a multi-country survey operations platform used across Nigeria, Zambia, and Botswana for PHIA HIV studies.

Technologies: C# ASP.NET, MSSQL, MySQL, AWS, OAuth2 (IdentityServer4).

Software Development Specialist

MGIC | Abuja, Nigeria

2017 – 2021

- Contributed to the development and nationwide deployment of NigeriaMRS (OpenMRS fork) used across government hospitals.
- Engineered core components of Nigeria Data Repository (NDR 3.0) supporting 1.9M+ HIV patients, optimizing XML ingestion with multiprocessing pipelines.
- Led development of STaRT, a laboratory sample tracking system piloted by the Malawi Ministry of Health.
- Built a logistics management platform used during the Population HIV Impact Assessment (PHIA) for laboratory supply coordination.

Technologies: C# ASP.NET, Java (Spring), MSSQL, MySQL.

Software Development Intern

MGIC | Abuja, Nigeria

2016 – 2017

- Designed and developed an HIV Testing Services (HTS) application used for national testing data capture and reporting.

- Built MSERS, a disease reporting and monitoring system for the Nigeria Centre for Disease Control (NCDC), enabling structured case reporting and surveillance data submission.
- Developed AMC, a drug consumption reporting tool supporting national pharmaceutical usage monitoring and regulatory reporting.

Technologies: C# ASP.NET, Java (Android), MSSQL.

CONSULTING & INDEPENDENT PROJECTS

Data Manager - Mpox Vaccine CEM Monitoring System

NAFDAC | Abuja, Nigeria

2025 – PRESENT (REMOTE)

- Designed and deployed a vaccine safety monitoring platform integrating CPro data collection with ASP.NET dashboards.
- Built automated ingestion services syncing surveillance data to WHO Vigilance Hub
- Implemented secure cloud deployment within government Azure infrastructure

ERP Software Consultant

RACIDA INTERNATIONAL | Nairobi, Kenya

2021 – 2022 (REMOTE)

- Architected a full ERP system supporting HR, procurement, inventory, vendor management, and performance tracking

Consultant Engineer CPro

APIN – Public Health Initiative | Abuja, Nigeria

2024 – 2025 (REMOTE)

- Designed and maintained CPro-based national survey data collection tools used for AIDS Impact Surveys.

EDUCATION

MBA, Management Information Systems

UNICAF University Zambia, 2021

B.Eng Mechanical Engineering

Federal University of Technology Minna, 2015